

Physics is the scientific study of matter, its fundamental constituents, its motion and behavior through space and time, and the related entities of energy and force.[1] It is one of the most fundamental scientific disciplines.[2][3][4] A scientist who specializes in the field of physics is called a physicist. Physics is one of the oldest academic disciplines.[5] Over much of the past two millennia, physics, chemistry, biology, and certain branches of mathematics were a part of natural philosophy, but during the Scientific Revolution in the 17th century, these natural sciences branched into separate research endeavors. Physics intersects with many interdisciplinary areas of research, such as biophysics and quantum chemistry, and the boundaries of physics are not rigidly defined. New ideas in physics often explain the fundamental mechanisms studied by other sciences[2] and suggest new avenues of research in these and other academic disciplines such as mathematics and philosophy. Advances in physics often enable new technologies. For example, advances in the understanding of electromagnetism, solid-state physics, and nuclear physics led directly to the development of technologies that have transformed modern society, such as television, computers, domestic appliances, and nuclear weapons;[2] advances in thermodynamics led to the development of industrialization; and advances in mechanics inspired the development of calculus.

History Main article: History of physics The word physics comes from the Latin *physica* ('study of nature'), which itself is a borrowing of the Greek *φυσική* (*phusikḗ* 'natural science'), a term derived from *φύσις* (*phúsis* 'origin, nature, property').[6][7][8]

Ancient astronomy Main article: History of astronomy Ancient Egyptian astronomy is evident in monuments like the ceiling of Senemut's tomb from the Eighteenth Dynasty of Egypt. Astronomy is one of the oldest natural sciences. Early civilizations dating before 3000 BCE, such as the Sumerians, ancient Egyptians, and the Indus Valley Civilization, had a predictive knowledge and a basic awareness of the motions of the Sun, Moon, and stars. The stars and planets, believed to represent gods, were often worshipped. While the explanations for the observed positions of the stars were often unscientific and lacking in evidence, these early observations laid the foundation for later astronomy, as the stars were found to traverse great circles across the sky,[5] which could not explain the positions of the planets. According to Asger Aaboe, the origins of Western astronomy can be found in Mesopotamia, and all Western efforts in the exact sciences are descended from late Babylonian astronomy.[9] Egyptian astronomers left monuments showing knowledge of the constellations and the motions of the celestial bodies,[10] while Greek poet Homer wrote of various celestial objects in his *Iliad* and *Odyssey*; later Greek astronomers provided names, which are still used today, for most constellations visible from the Northern Hemisphere.[11]

Natural philosophy Main article: Natural philosophy Natural philosophy has its origins in Greece during the Archaic period (650 BCE – 480 BCE), when pre-Socratic philosophers like Thales rejected non-naturalistic explanations for natural phenomena and proclaimed that every event had a natural cause.[12] They proposed ideas verified by reason and observation, and many of their hypotheses

proved successful in experiment;[13] for example, atomism was found to be correct approximately 2000 years after it was proposed by Leucippus and his pupil Democritus.[14] Aristotle and Hellenistic physics